

**Course Outline Draft of  
Pattern Recognition  
Semester: Spring 2026**

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**Course Description:**

Introduction to pattern recognition: features, classifications, learning. Statistical methods, structural methods, and hybrid method. Applications to speech recognition, remote sensing, and biomedical area. Learning algorithms, Syntactic approach: Introduction to pattern grammars and languages. Parsing techniques. Pattern recognition in computer-aided design. Introduction to formal languages, String languages for pattern description, Higher dimensional pattern grammars, Syntax analysis as a recognition procedure, Stochastic languages, Error-corresponding parsing for string languages, Error-correcting tree automata, cluster analysis for syntactic patterns, Grammatical inference for syntactic approach to texture analysis.

**Prerequisites and Co-requisites:** Background in Programming, Probability, Statistics, Advanced AI and Machine Learning is helpful but not required..

**Course Outcomes:**

|      |                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------|
| CO 1 | <b>Acquire</b> introductory concepts that are relevant to pattern recognition techniques.               |
| CO 2 | <b>Analyze</b> and understand algorithms and related statistics to be able to analyze patterns in data. |
| CO 3 | <b>Comprehension</b> of current research on pattern recognition and futuristic pattern evolutions.      |

**Class Organization:** Semi-flipped learning environment.

**Lecture Plan, Tentative**

| Week      | Topic                                                                                                                                                                                                                               | Notes and Readings                                                                                                           | Quiz                |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1.        | Pattern, Probability, and Statistical Learning<br>Challenges in Naive Bayes                                                                                                                                                         | Lecture notes on intro, NB<br>PRML Ch <a href="#">1</a> , <a href="#">2</a><br>ISLP Ch <a href="#">1</a> , <a href="#">2</a> |                     |
| 2.        | Pattern, Probability, and Statistical Learning<br>Challenges in Naive Bayes, <i>Network</i><br>(continued)<br><i>High-Dimensional Statistics</i><br>Linear and Logistic Regression<br><i>Statistical Thinking for Public Health</i> | Lecture notes on NB, BN<br>PRML Ch <a href="#">3</a> , <a href="#">8</a><br>ISLP Ch <a href="#">3</a><br>SLP3 Ch<br>ESL Ch 3 |                     |
| 3.        | Classification<br>Ethics, Bias and Real-world Pattern<br>Limitations<br>KNN limits and optimizations                                                                                                                                | Lecture notes on Ethics, Bias,<br>KNN<br>PRML Ch 4<br>ISLP Ch <a href="#">4</a><br>ESL Ch 13                                 |                     |
| 4.        | Challenges of Support Vector Machine (SVM)                                                                                                                                                                                          | Lecture notes on SVM<br>PRML Ch 6, 7<br>ISLP Ch <a href="#">9</a><br>ESL Ch 12                                               | Q1 on Feb 25, WED   |
| 5.        | Unsupervised Learning, Kmeans, Hierarchical<br>Clustering                                                                                                                                                                           | Lecture notes on Kmeans, HC<br>ISLP Ch <a href="#">12</a><br>ESL Ch 14                                                       | Q2 on Mar 2, MON    |
| <b>6.</b> | <b>Midterm</b>                                                                                                                                                                                                                      | <b>On Sl 1 to 5</b>                                                                                                          |                     |
| 7.        | Resampling Methods,<br>Linear Model Selection and Regularization                                                                                                                                                                    | ISLP Ch <a href="#">5</a> ,<br>PRML Ch 5<br>ISLP Ch <a href="#">6</a><br>ESL Ch 4, 5, 7                                      | Q3 on Mar 11, WED   |
| 8.        | Moving Beyond Linearity                                                                                                                                                                                                             | PRML Ch 12<br>ISLP Ch <a href="#">7</a>                                                                                      |                     |
| 9.        | Tree-Based Methods                                                                                                                                                                                                                  | PRML Ch 14<br>ISLP Ch <a href="#">8</a><br>ESL Ch 9                                                                          |                     |
| 10.       | Hybrid Approaches: SyntPR with Statistical,<br>Structural, and Neural approaches: Deep<br>Learning, Transformers, Attention,                                                                                                        | PRML Ch 5<br>ISLP Ch <a href="#">10</a>                                                                                      | Q4 on April 20, MON |

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|-----|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--|
|     | Geospatial Data Patterns<br>Astronomical Data Patterns<br>Public Health Data Patterns<br>MRI Data Patterns                 |                                                                                      |  |
| 11. | Survival Analysis and Censored Data                                                                                        | ISLP Ch <a href="#">11</a>                                                           |  |
| 12. | Multiple Testing<br>Hidden Markov Model (HMM, tHMM)<br>Gaussian Mixture Model (GMM)                                        | ISLP Ch <a href="#">13</a><br>Lecture notes on GMM<br>PRML Ch 9, 10, 13<br>ESL Ch 13 |  |
| 13. | Graph Grammar applications, GNN,<br>Knowledge Graphs, Social Network Analysis                                              |                                                                                      |  |
| 14  | Evolution of Multimodal Data and Patterns:<br>Challenges and Applications, Open research<br>problems and future directions |                                                                                      |  |

### Recommended Readings

1. Pattern Recognition and Machine Learning (PRML), Christopher M. Bishop, Springer, <http://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning>
2. Syntactic Pattern Recognition, Mariusz Flasiński, World Scientific
3. Applications of Pattern Recognition, King Sun Fu, CRC Press
4. Pattern Recognition: Statistical, Structural and Neural Approaches, Robert J Schalkoff, Wiley
5. An Introduction to Statistical Learning (ISLP), Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Springer, <http://www.statlearning.com>
6. The Elements of Statistical Learning (ESL), Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer, <http://www-stat.stanford.edu/ElemStatLearn>

### Supplementary Readings

1. Pattern Classification, Richard O. Duda, Peter E. Hart, David G. Stork, Wiley
2. Understanding Machine Learning, Shai Shalev-Shwartz and Shai Ben-David, Cambridge University Press
3. Syntactic Pattern Recognition for Seismic Oil Exploration, Kou-Yuan Huang, World Scientific
4. Syntactic and Structural Pattern Recognition, Gabriel Ferraté, Springer-Verlag Berlin Heidelberg
5. Syntactic Pattern Recognition and Applications, King Sun Fu, Prentice-Hall
6. Syntactic Methods in Pattern Recognition, King Sun Fu, Academic Press
7. Handwriting Analysis through Syntactic Pattern Recognition, Ashok Jammi, LAP LAMBERT

### Others.

Domain specific books where syntactic pattern recognition is to be applied and proceedings from related journals, conferences, and workshops such as

1. S+SSPR: Joint IAPR International Workshops on Statistical Techniques in Pattern Recognition (SPR) and Structural and Syntactic Pattern Recognition (SSPR), <https://link.springer.com/conference/sspr>
2. Pattern Recognition, Elsevier, <https://www.sciencedirect.com/journal/pattern-recognition>
3. Pattern Recognition Letters, Elsevier, <https://www.sciencedirect.com/journal/pattern-recognition-letters>
4. Advances in Computer Vision and Pattern Recognition, Springer <https://www.springer.com/series/4205>

Other books on Pattern Recognition, Formal Languages, can be referred to, as necessary

### Supporting Tools:

- buX, <http://bux.bracu.ac.bd>
- Google Classroom

### Course Policies:

- Regular participation is required via class and other discussion channels. Students are expected to engage in discussions, reply, and post materials they have explored or learned from the broader domain of the subject.
- You should take detailed notes. If you face difficulties in understanding, or if you think there are corrections, please notify!
- Please respect your classmates, class representatives, and instructor by preventing distractions:
  - o Please pay attention to the instructor and participate in the discussions. Please postpone side conversations to the break / after class.
  - o Please do not do other work during classes.

Honor Code: Any form of cheating, plagiarism, or academic dishonesty will result in an “F” grade in the course.

Late Work and Examinations: Students who know that they are going to miss class should make arrangements in advance. There will not be any make-up for quizzes, except the cases of hospitalization or detention.

**Grading Policies:**

Student's grades are assigned according to the grading scale of the Brac University and Examinations Regulations. In addition, the faculty are allowed to take into consideration the class average and standard deviation to reflect the actual class performance for student grade assignment. The grades at the university will be indicated in the following manner:

| Marks   | Grades     |
|---------|------------|
| 97-100  | A+ (4.0)   |
| 90- <97 | A (4.0)    |
| 85- <90 | A- (3.7)   |
| 80- <85 | B+ (3.3)   |
| 75- <80 | B (3.0)    |
| 70- <75 | B- (2.7)   |
| 65- <70 | C+ (2.3)   |
| 60- <65 | C (2.0)    |
| 57- <60 | C- (1.7)   |
| 55- <57 | D+ (1.3)   |
| 52- <55 | D (1.0)    |
| 50- <52 | D- (0.7)   |
| <50     | F (0.0)    |
| P       | Pass       |
| I       | Incomplete |
| W       | Withdrawal |
| R       | Retaken    |

**Course Assessment Methods: (tentative)**

| Section                | Marks (%)   |
|------------------------|-------------|
| Quiz                   | 20%         |
| Concept Paper          | 10%         |
| Hands-on Exploration   | 10%         |
| Midterm                | 20%         |
| Proposal               | 10%         |
| Project Implementation | 5%          |
| Presentation           | 10%         |
| Final Report           | 15%         |
| <b>Total</b>           | <b>100%</b> |

**Date:** January 28, 2026

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